* Type Casting in Java :-

=> Java doesn't allow automatic conversion becausing it wants to protect you from data loss.

Ex :-

```
float a = 25.0;
int b = a;
```

gives error

assigned float in int

not automatic conversion

```
int b = (int) a;
```

↳ typecast float a in to int

↳ then assign to int b

data loss of .0

=> Typecasting is known as "Narrowing Conversion" / "Explicit conversion"

```
public class javaBasics {
    public static void main(string args[]) {
        char ch = 'a';
        char ch2 = 'b';
        int num1 = ch;
        int num2 = ch2;

        System.out.println (num1);
        System.out.println (num2);
    }
}
```

97

98

⇒ in java typecasting of char is also possible

a, b, c, d, e, f, - - - - - -

97 98 99 100 101 102 - - - -

* Type Promotion in Expressions :-

1.) Java automatically promotes each byte, short, or char operand to int when evaluating an expression.

```
char a = 'a';
short b = 50;
```

char ← a + b → short

Java automatically convert to int

↓

then calculate sum

Ex:- pseudocode

```
char a = 'a';
char b = 'b';

System.out.println(a);
System.out.println(b - a);
```

98 97

print → a

print → (98-97) = 1

↓

⇒ Java only promotes conversion in only Expression.

Ex:-

char c = a - b; → Error

$\underbrace{\text{char c}}_{\text{char}} = \underbrace{a - b}_{\text{Integer}}$

2.) if one operand is long, float, or double the whole expression is promoted to long, float, or double respectively.

⇓

if largest possible data type is long

⇓

all converted to long

(similarly for float or double)

Ex:-

int a = 10;
float b = 20.25f;
long c = 25;
double d = 30;

convert whole expression into double

int float long double

double ans = a + b + c + d;

$\underbrace{\text{int ans}}_{\text{int}} = \underbrace{a + b + c + d}_{\text{double}}$; ✗

Error

Ex:- byte b = 5;

b = b * 2;

byte a = $\underbrace{\underbrace{b}_{byte} * 2}_{Expression}$;

Expression
↓
automatically convert to int

b * 2

5 * 2 = 10

↳ is 'int' not byte

↳ but 10 lies in the range of byte

-128 to 127

↳ despite java thought that some information might be lost

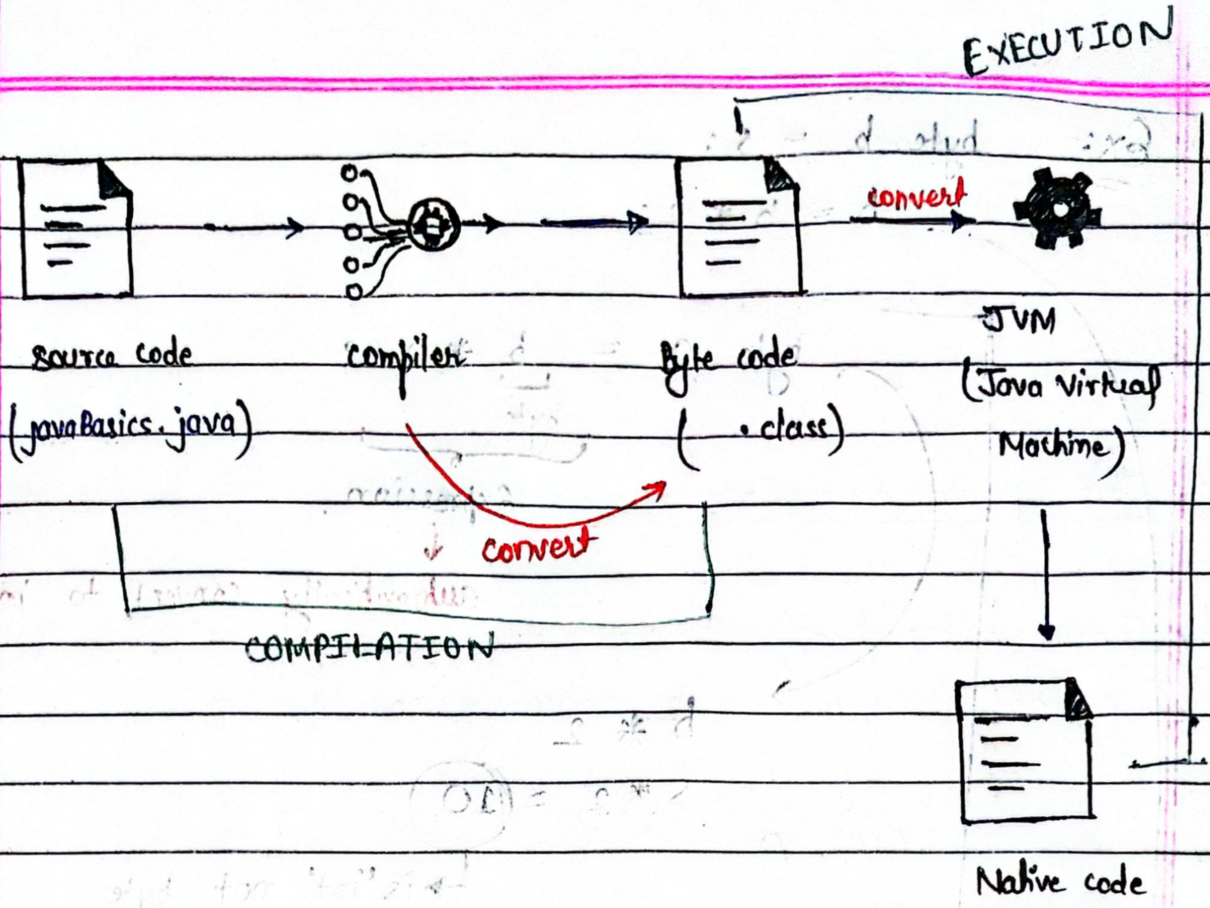
b = (byte) (b * 2); → Typecasting

* How is your code running :-

JDK → Java development kit

JRE → Java Runtime Environment

JVM + libraries + development tools



★ windows have different Native code while mac have different and also Linux has different Native code

⇓

you can write java code in any machine

⇓

So, Java is "PORTABLE LANGUAGE".

⇒ c++ is not portable language.